

Mirroring the Scene Flips the Score: Matched Controls for Directional Language Grounding

Ali-Adeeb Abbas^{1,2}, Esmail Seraj², Behrad Toghi² and Taskin Padir^{1,3}

Abstract—Binary success gaps between opposite spatial commands are often treated as direct evidence of language grounding, even though completion also depends on scene geometry. For $\pi_{0.5}$, reflecting only the movable-object centers reverses completion: 4/27 LEFT and 25/27 RIGHT become 25/27 and 9/27, with the policy, prompts, controller, and seeds unchanged. The near-mirror pairing shows that the measured gap belongs to the policy–layout pair, not to the checkpoint alone. Matched reference inversion exposes a second limit: describing the same physical goals from the other object’s frame collapses $\pi_{0.5}$ endpoint separation from 23.28 cm to 0.19 cm. A binary gap can also conceal placement change: in Nano’s matched 27-seed symmetric-scene core, the binary interaction is exactly zero ($p = 1$) alongside a -13.9 cm requested-side-depth interaction. A powered $\pi_{0.5}$ residual remains and fails the prespecified equivalence criteria; the symmetric-scene control therefore neither removes every disparity nor localizes the residual to the policy. Together, these results show why neither a large nor a small completion gap identifies directional grounding without matched tests of wording and layout. We recommend reporting prompt-conditioned action response, signed endpoint ordering, requested-side depth, and frozen task completion separately, together with matched language and scene controls.

Index Terms—Vision-language-action models, robot learning, manipulation, language grounding, evaluation methodology.

I. INTRODUCTION

Vision-language-action (VLA) policies promise a single interface for open-ended robot control: a user states a goal in words, and one pretrained model produces the actions [1]–[4]. Whether that interface delivers is an empirical question, and relational placement has become a standard way to ask it. A representative protocol holds the scene fixed and asks the policy to place an object in “the pot on the left” or “the pot on the right.” It balances trials between the two directions and scores each with a frozen success rubric [5]. It is tempting to read a large LEFT/RIGHT gap as a language-grounding failure: the model understood one direction but not the other, or vision overrode language [6].

That reading is plausible because the failure it names is real. Robot datasets carry formulaic, low-diversity instruction labels, making the instruction largely predictable from the observation [7]. A conditioned policy can then ignore its language input, the posterior collapse familiar from conditional sequence models [8], [9]. It also motivates the remedies:

counterfactual relabeling [10], denser command interfaces [5], [11], [12], and test-time language steering [13]. These methods address genuine weaknesses in instruction following. The attribution problem is whether a binary directional score shows that weakness in a particular experiment [14].

That score combines several questions. For a trial to succeed, the instruction has to alter control, the change has to point toward the requested relation, the robot has to execute a feasible grasp and transport, and the final state has to cross a binary task boundary. A policy can pass the first two steps but fail the last two because one side of a particular layout is harder to reach. The reverse mistake is also possible: a policy can react to the literal tokens LEFT and RIGHT without preserving the physical goal once the linguistic reference frame is inverted [15]. A single binary contrast can therefore produce two opposite errors: a false failure when language redirects the endpoint but geometry blocks completion, and a false pass when familiar wording elicits the expected behavior without relation-level transfer.

Our central finding is an invariance failure: the directional gap is a property of the policy–layout pair, not a layout-invariant property of the checkpoint. Reflection reverses completion for $\pi_{0.5}$: 4/27 LEFT and 25/27 RIGHT successes become 25/27 and 9/27 after the movable-object centers are mirrored, with the policy, prompts, controller, and seeds unchanged (Fig. 1). The near-mirror pairing is striking: reflected LEFT exactly matches control RIGHT at 25/27, while reflected RIGHT is 9/27 against control LEFT at 4/27. We treat that five-success mismatch as the part not removed by this object-position reflection, without assigning it to a particular model-side mechanism.

Geometry is not the only threat to interpretation. When the same physical goals are expressed from the other object’s reference frame—“cube left of bowl” becomes “bowl right of cube,” and conversely— $\pi_{0.5}$ endpoint separation collapses from 23.28 cm to 0.19 cm. Base-frame sensitivity to LEFT and RIGHT therefore does not establish that the underlying relation transfers across linguistic frames.

The contribution is not merely an additional metric; it changes what a directional score can support. Before intervention, the $\pi_{0.5}$ completion gap is consistent with asymmetric grounding. After reflection, the same evaluation favors the opposite direction, even though the prompt still redirects the endpoint. Reference inversion exposes the complementary error: a reassuring base-frame response need not survive an equivalent statement of the physical relation. Together, the controls show

¹RIVeR Laboratory, Northeastern University, Boston, MA, USA. abbas.a@northeastern.edu

²General Motors, Mountain View, CA, USA.

³Amazon Robotics, North Reading, MA, USA.

why neither a large nor a small binary gap identifies grounding without matched interventions on both wording and layout.

We make three contributions, ordered by the claims they support:

- **A matched object-layout intervention that reverses directional completion.** A pure reflection reverses $\pi_{0.5}$ completion. It also reverses the requested-side-depth advantage in three DROID/RoboLab checkpoints; graded displacement and symmetric-scene controls show the same dependence in continuous placement.
- **A reference-inversion control for relation-level transfer.** Holding each physical goal and the manipulated object fixed while changing the linguistic frame reduces $\pi_{0.5}$ endpoint separation from 23.28 cm to 0.19 cm.
- **An evidence-gated directional evaluation protocol.** We separate executed-action response, signed endpoint ordering, requested-side depth, and frozen task completion. Scene effects are interpreted only where endpoint redirection survives; Nano’s matched 27-seed symmetric-scene core has zero binary interaction alongside a -13.9 cm depth interaction, showing why this decomposition is necessary.

None requires retraining, new hardware, or a change to the frozen success predicate.

II. RELATED WORK

Generalist language-conditioned policies. Modern manipulation policies pair pretrained visual-language representations with expressive action decoders. Representative systems include π_0 and $\pi_{0.5}$, which use flow-matching action models, OpenVLA, an open autoregressive alternative, and RT-2, which transfers web knowledge to control [1]–[4]. Hierarchical variants place a reasoning model above the controller and pass it natural-language subtasks [16]. Scale has expanded the tasks one policy can address, but it also makes a failed episode difficult to attribute: language interpretation, object localization, grasping, transport, release, and task scoring all lie on the same path to success.

Steerability and instruction following. A growing literature treats weak instruction following as the bottleneck and enriches the command interface. CAST augments datasets with counterfactual language [10], while STEER relabels demonstrations to expose a low-level control interface [11]. Steerable Policies train on subtask, motion, trace, and point commands [5], and InSight makes individual primitives addressable [12]. Other work steers a frozen policy at test time [13] or asks how steerability survives post-training [14], [17]. These methods address genuine failures of instruction following, but they rely on directional and relational success contrasts to certify that the failure was linguistic in the first place.

Diagnosing whether language is used. Counterfactual evaluation reduces attribution ambiguity by holding the visual input fixed while changing the instruction. LIBERO-CF and Counterfactual Action Guidance examine cases in which visual evidence dominates a conflicting instruction [6], and recent

analyses bound how far instruction following transfers to unseen phrasings [15]. Steering protocols also report graded task progression, per-primitive progress, or success by perturbation condition [5], [12], [13]. Those approaches grade progress toward the command that was issued. Our contrast asks a different question: when two commands share the same reset and policy-sampling seed, what changes because one relation word changes? The exact pairing requires a resettable simulator, but it prevents scene and sampling differences from entering the prompt contrast [12].

Evaluation practice. Grounding spatial and directional language into executable behavior long predates VLAs, and early systems already separated the referring expression, the spatial relation, and the executed path [18], [19]. Recent large-scale evaluations, including the TRI study of Large Behavior Models, strengthen manipulation comparisons through controlled initial conditions, randomized trials, and statistical confidence [20]. We share that emphasis on controlled evaluation but target a different estimand: rather than aggregate success or robustness across tasks, we contrast two relational commands at the same reset and sampling seed. DROID and RoboTwin illustrate the distinct data and simulation ecosystems in which such policies are trained and tested [21], [22]. A frozen binary predicate answers whether the task was completed in the evaluated scene distribution, but it can saturate at ceiling, collapse at floor, or hide large continuous placement changes. We keep it as the final outcome and pair it with signed endpoint and requested-depth measurements and with interventions that vary language form and scene geometry separately.

III. MATCHED AUDIT DESIGN

The audit layers two controls on the same base-frame prompt pair. Reference inversion changes the linguistic frame while holding each physical goal fixed; the scene interventions change object geometry while preserving the prompt pair. Together they test whether a directional score is stable to either representation of the task.

Minimal-pair token intervention. Literal prompt tokens appear in small capitals, LEFT and RIGHT, while *leftward* and *rightward* denote physical directions in robot coordinates; a policy can track the token without preserving the direction it denotes. For each reset s_i we run a policy under prompts ℓ_i^L and ℓ_i^R that differ only in one opposite direction word, holding scene state, checkpoint, controller, environment seed, policy-sampling seed, action interface, and horizon fixed. Both prompts therefore execute from the same initial state and random draws; the direction word is the designed difference. We issue the instruction once at reset and leave it alone: no oracle actions, no prompt switching, and no language conditioned on progress. DROID/RoboLab pairs the DROID Franka embodiment [21] with the RoboLab Isaac Sim environment [23] and uses a Rubik’s cube and bowl; RoboTwin [22] uses a wooden block and playing-card box. Both require placement and release in the requested relational region.

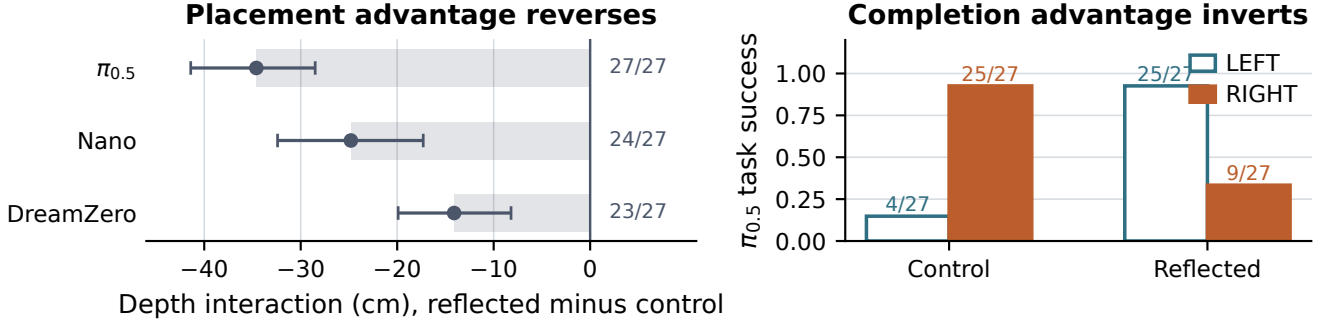


Fig. 1. **Reflecting movable-object positions reverses placement advantage within checkpoint.** Left: reflected-minus-control interaction in requested-side depth, with seed-clustered 95% bootstrap intervals; fractions give seeds with a negative interaction. Right: $\pi_{0.5}$ completion changes from 4/27 LEFT and 25/27 RIGHT in control to 25/27 and 9/27 after reflection. Reflected LEFT exactly matches control RIGHT, while reflected RIGHT approaches control LEFT. All causal comparisons are within checkpoint; the panels do not support cross-model ranking.

TABLE I
EVALUATED CHECKPOINT–ARENA ROWS; SHORT NAMES IN PARENTHESES.

Model (short name)	Class	Pairs
<i>DROID/RoboLab: cube to the requested side of a bowl</i>		
$\pi_{0.5}$ DROID [4]	VLA	27
GR00T N1.7 DROID (GR00T) [25]	VLA	27
Cosmos3 Edge Policy DROID (Edge) [24]	WAM	27
Cosmos3 Nano Policy DROID (Nano) [24]	WAM	27
DreamZero DROID (DreamZero) [26]	WAM	27
<i>RoboTwin: block to the requested side of a card box</i>		
Efficient-WAM-RT [27]	WAM	63
FastWAM [28]	WAM	63
LingBot-VA [29]	WAM	63

Evaluation corpus. Table I lists the evaluated checkpoints; App. B records the available artifact identifier and revision behind each row. Two are action-only VLAs and six are world-action models (WAM), which predict future observations alongside actions. Within each arena, checkpoints use its fixed action interface and frozen predicate. Edge and Nano are the 4B and 16B policies of one Cosmos3 family [24], so the rows are not fully independent. The screen runs one matched-pair row per checkpoint and arena before any intervention: five DROID/RoboLab checkpoints at 27 matched pairs each and three RoboTwin checkpoints at 63, for 324 pairs, or 648 episodes. The two arenas use different tasks and different success tests, so no outcome estimate or effect pools arenas; totals across rows are bookkeeping only, and the larger intervention cohorts state their own denominators.

Behavioral measurements. Let $a_{1:T_L}^L$ and $a_{1:T_R}^R$ be the executed action sequences in a minimal pair, and let y_i^L, y_i^R be the final target-minus-reference lateral offsets in robot coordinates, where positive is robot-left. The four measurements correspond to four stages of the task: prompt response, directional ordering,

placement quality, and completion. We report

$$A_i = \mathbb{K}[a_{1:T}^L \neq a_{1:T}^R], \quad T = \min(T_L, T_R), \quad (1)$$

$$D_i = y_i^L - y_i^R, \quad (2)$$

$$S_i^d = \mathbb{K}[\text{frozen task predicate is satisfied for } d], \quad (3)$$

$$M_i^L = y_i^L, \quad M_i^R = -y_i^R. \quad (4)$$

A_i uses bitwise inequality on the complete common executed prefix. Positive D_i means the requested endpoint ordering: within a pair the object came to rest further leftward under LEFT than under RIGHT. S_i^d is task completion, and requested-side depth M_i^d records where the target comes to rest along the requested physical direction.

Calibrating A_i against sampling noise. A_i tests response, not semantics, and alone it is weak evidence: stochastic sampling makes two executions differ even under an unchanged prompt. We therefore calibrate it on a single frozen observation, querying the policy without executing the robot so that no difference compounds through dynamics. The diagnostic issues 336 action requests across six conditions (three DROID/RoboLab checkpoints crossed with the control and reflected scenes), each including a repeated request that must return identical output. Its summary is a prompt-to-noise ratio: the LEFT/RIGHT action difference at a matched seed over the root-mean-square difference between distinct seeds under an unchanged prompt. When repeated requests are bit-identical in this harness, the denominator is zero; for those rows we report the numerator alone rather than treating an undefined ratio as a small one.

Reference inversion. Base-frame prompts change the surface token and physical goal together; the reference-inversion control instead holds the physical relation fixed and changes its linguistic frame. The base-frame prompts are “Put the Rubik’s cube to the left of the bowl.” and “Put the Rubik’s cube to the right of the bowl.” The inverted prompts are “Place the Rubik’s cube so that the bowl is to the right of the Rubik’s cube.” and the same sentence with *left*. All four name the cube as the object to place, so inversion changes the frame of the relational clause and not the manipulandum; the direction word flips while the intended physical relation does not. The scorer,

and the sign of M for this control, follow the physical goal rather than the prompt text. The $\pi_{0.5}$ study uses 341 resets in one symmetric DROID/RoboLab scene, with both forms of both physical goals sharing reset and sampling state, for 1,364 episodes on seeds disjoint from every scene-intervention cohort.

Geometry interventions. Reflection is a pure world-space object-position intervention fixed in advance: it mirrors the centers of all movable objects about the robot sagittal plane without mirroring observations or un-mirroring actions. The robot, cameras, fixed scene geometry, prompts, seeds, and controller remain unchanged. We pair it with two further interventions fixed in advance: a seven-level sweep that moves only the reference object laterally and a symmetric-scene cohort that drives a registered movable-object asymmetry metric to zero (App. A). The symmetric package also changes which other objects are present, so it is not a position-only intervention and does not imply bilateral embodiment symmetry. Scene-effect claims are made only for rows that pass the prespecified endpoint-redirection positive control, which asks whether the policy still moves its endpoint in the commanded direction. Failed rows are reported but remain uninterpretable for attenuation of a directional gap.

Scoring and inference. We do not relax the original task predicate. Each episode receives exactly one outcome label, assigned in a fixed order: pick failed, transport failed, wrong side, release failed, or correct. Every behavioral failure stays in its denominator. Marginal binary rates use Wilson intervals, paired continuous contrasts use seed-clustered percentile bootstraps with 20,000 resamples and exact sign tests, and layout interactions use exact within-seed label permutations. Absence claims require two one-sided equivalence tests (TOST) against prespecified margins, rather than nonsignificance [30]–[32]. Prompts, seeds, success tests, target quantities, margins, and the failure-label order are all fixed before the runs they affect. The public audit archive¹ records the available manifests, seeds, checkpoint revisions, and freeze times for the reported cohorts.

IV. REFERENCE INVERSION COLLAPSES ENDPOINT SEPARATION

Reference inversion collapses endpoint separation for $\pi_{0.5}$ from 23.28 cm to 0.19 cm on matched physical goals (Fig. 2). This is the linguistic counterpart to the scene controls: a familiar base-frame response does not survive when the same relation is expressed from the other object’s perspective.

Base-frame positive control. All 324 base-frame minimal pairs across the eight rows execute different action sequences: 135 DROID/RoboLab pairs and 189 RoboTwin pairs. Endpoint ordering is positive in 125/135 and 128/189 pairs, respectively. These bitwise differences supply the matched trace-difference positive control. They do not show that the policy represents the relation independently of its wording, nor do they guarantee task completion.

The fixed-observation diagnostic of Sec. III separates that response from sampling noise. For $\pi_{0.5}$, prompt-induced action difference was about 4% of cross-seed same-prompt variation, yet the difference vectors shared a consistent orientation across seeds (mean cosine 0.540 in control and 0.652 after reflection; permutation $p < 10^{-5}$). For Nano the same magnitude ratio was 0.72 under the control scene and 1.19 after reflection, but the orientations were far less consistent (mean cosine 0.027 and 0.079; control permutation $p = .167$). Repeated DreamZero requests were bit-identical in this fixed-observation harness, so its ratio is undefined rather than small; whether that behavior belongs to the checkpoint or to harness seeding remains unresolved. These open-loop rankings did not match closed-loop endpoint ordering, so action difference is a positive control, not a standalone grounding score. A small, consistently oriented change could accumulate during replanning, but that explanation remains untested.

Matched reference inversion. Reference inversion asks whether the same physical relation survives being expressed from the other object’s frame. Under inversion the direction word and the physical goal point opposite ways, which separates two behaviors that base-frame prompts cannot tell apart. If the policy tracks the relation, the physical-LEFT-minus-physical-RIGHT endpoint contrast should be positive. If it tracks the direction word regardless of the sentence frame, the contrast should be negative, reaching -23.28 cm if the base-frame response transferred unchanged.

Base-frame prompts separate the endpoints by $+23.28$ cm (95% CI [21.89, 24.56]), whereas reference-inverted prompts yield only $+0.19$ cm ($[-0.88, 1.27]$). The observed response therefore collapses rather than transferring at the base-frame scale by either rule; the inverted interval includes zero, although equivalence to zero was not established. Out-of-distribution phrasing remains an alternative explanation that this experiment does not separate from failed relation transfer. This conclusion is specific to endpoint direction, not all prompt sensitivity; all 682 base-frame/inverted physical-goal pairs execute different action traces.

Inversion also reduces placement depth within each physical goal. The inverse-minus-base-frame change is -9.78 cm for physical LEFT (90% CI $[-10.92, -8.59]$) and -13.31 cm for physical RIGHT ($[-14.07, -12.54]$); both intervals lie outside the prespecified ± 4.15 cm equivalence margin. Success falls from 210/341 to 151/341 for LEFT and from 297/341 to 240/341 for RIGHT. These rates make a wholesale switch to moving only the bowl less consistent with the evidence: the predicate scores placement of the cube, yet 151 and 240 of 341 inverted episodes still satisfy it. Because reference-object displacement was not measured, the counts do not exclude incidental bowl motion or a mixed strategy. The corresponding changes, -17.30 and -16.72 percentage points, do not meet the prespecified ± 15.56 -point equivalence criterion. Base-frame response is therefore insufficient evidence that a physical relation is preserved across reference frames.

¹<https://github.com/adeeb10abbas/steerable>

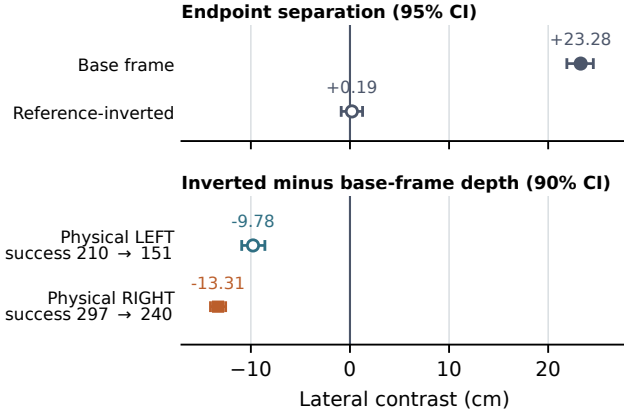


Fig. 2. **Reference inversion collapses directional endpoint separation for $\pi_{0.5}$.** Top: physical-LEFT-minus-physical-RIGHT endpoint separation with seed-clustered 95% bootstrap intervals; positive is relation-consistent. Bottom: inverted-minus-base-frame requested-side depth with 90% intervals; labels give base-frame-to-inverted success counts out of 341. This cohort’s seeds are disjoint from those of the scene-intervention experiments.

V. REFLECTION REVERSES DIRECTIONAL COMPLETION

Reflection reverses completion for $\pi_{0.5}$: 4/27 LEFT and 25/27 RIGHT successes become 25/27 and 9/27 after only the movable-object centers are mirrored in world coordinates, with the policy, prompts, controller, and seeds unchanged. This is a pure world-state intervention: only movable-object centers are reflected; observations are rendered normally and actions are executed without a coordinate transform. The resulting pairing is nearly mirrored. Control RIGHT and reflected LEFT each succeed in 25/27 trials, while control LEFT and reflected RIGHT succeed in 4/27 and 9/27. Reflected RIGHT therefore has five additional successes relative to control LEFT; this is the descriptive remainder not removed by object-position reflection, not evidence for a particular policy-side mechanism.

The eight-row pre-intervention screen provides context rather than causal evidence (Fig. 3). $\pi_{0.5}$ and DreamZero favor RIGHT, Nano is near ceiling, and GR00T completes only 3/27 LEFT and 0/27 RIGHT trials; 49/54 GR00T episodes fail at pickup. The RoboTwin screening gaps are descriptively small, but two checkpoints later fail the endpoint-redirection positive control, so those nulls are uninterpretable rather than failed cross-arena replications.

A. Pure object-position reflection

Reflecting the centers of movable objects about the sagittal plane (Sec. III) gives four conditions per checkpoint (two directions by two layouts) over 27 seeds, or 108 episodes for each of three DROID/RoboLab checkpoints. The $\pi_{0.5}$ reflection cohort uses seeds disjoint from the screen, which is why Fig. 3 reports 5/27 and 24/27 where the control condition below has 4/27 and 25/27. The primary outcome is the within-seed layout interaction in requested-side depth; success is secondary because binary outcomes can saturate by layout and direction. Nano, the checkpoint closest to the binary ceiling, illustrates why in Fig. 4.

The reflected-minus-control requested-side-depth interaction is negative in every reflection checkpoint (Fig. 1). The interaction is -34.6 cm for $\pi_{0.5}$ (95% CI $[-41.4, -28.5]$; 27/27 seeds; $p = 1.49 \times 10^{-8}$). It is -24.8 cm for Nano ($[-32.4, -17.3]$; 24/27; $p = 4.92 \times 10^{-5}$) and -14.1 cm for DreamZero ($[-19.9, -8.2]$; 23/27; $p = 3.11 \times 10^{-4}$). For Nano and $\pi_{0.5}$, the estimated interactions in endpoint redirection are much smaller, $+0.5$ cm and -1.12 cm. Both carry $p = .701$ because the exact sign test sees mirrored 15/12 and 12/15 splits over the same 27 seeds, not because the estimates agree; equivalence was not established, so we do not call redirection invariant. The supported contrast is that placement changes far more than the estimated redirection interaction.

Each seed contributes $(Y_R^{\text{refl}} - Y_L^{\text{refl}}) - (Y_R^{\text{ctrl}} - Y_L^{\text{ctrl}})$, so a negative value means reflection reduces or reverses the control-layout RIGHT advantage. The mean of this per-seed completion interaction is -1.37 on its $[-2, 2]$ scale ($p = 5.96 \times 10^{-8}$). In this $\pi_{0.5}$ DROID/RoboLab cohort, reflecting the movable-object centers flips the observed completion advantage under the unchanged frozen predicate.

B. A graded intervention and converging controls

Reflection is a large intervention, so we also move the bowl across seven lateral positions while retaining the same Nano checkpoint and 15 matched seeds per level (210 episodes). Across all seven registered positions, the requested-depth contrast has a slope of 1.12 cm of depth per cm of reference displacement (95% CI $[0.72, 1.56]$); 13 of 15 seed-level slopes are positive (exact $p = .00739$). A post hoc middle-five sensitivity is weaker: 0.49 (95% CI $[-0.04, 0.97]$), with 11 of 15 positive slopes ($p = .118$). We therefore claim a graded dependence across the seven tested positions, not an interior linear law. Because the fitted zero crossing lies outside the tested support, the data do not identify a balanced interior layout.

Fig. 5 also shows why binary success alone is a weak diagnostic here: LEFT completion changes from 15/15 to 10/15, while RIGHT stays close to ceiling. Separate registered cohorts point in the same direction without being pooled with reflection. For Nano, moving the target’s start side changes success from 26/27 versus 22/27 to 23/27 versus 27/27 (interaction $+0.296$, $p = .0156$). Also for Nano, swapping target and reference roles changes the depth contrast by $+13$ cm (95% CI $[6, 21]$, $p = .00151$). The role swap keeps the direction word and relation template fixed but changes the manipulandum and physical goal; reference inversion instead keeps both fixed while changing the linguistic frame. For $\pi_{0.5}$, repeating policy sampling eight times in each of 27 fixed scenes gives 216 matched pairs per direction, with 41/216 LEFT and 197/216 RIGHT successes, showing that the gap is not an artifact of one draw.

VI. BINARY CEILINGS HIDE PLACEMENT CHANGE

Ceilings hide placement change: in the matched 27-seed symmetric-scene core, Nano has exactly zero binary interaction ($p = 1$) alongside a -13.9 cm requested-side-depth interaction

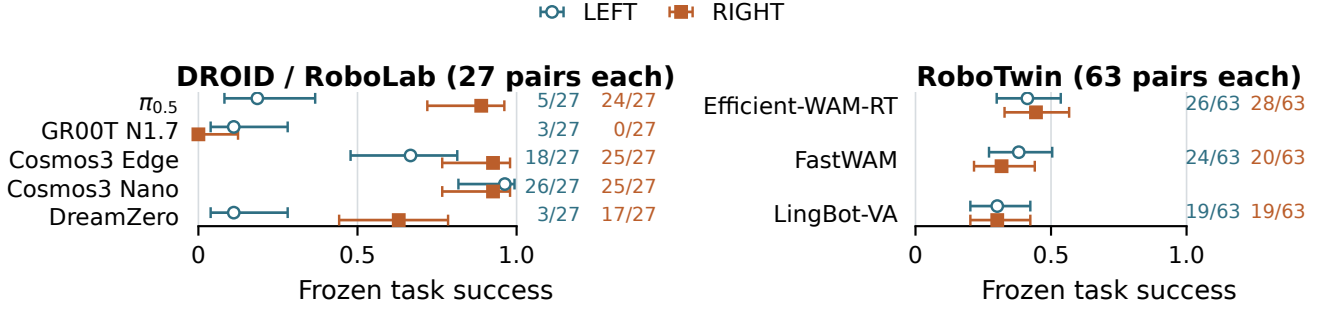


Fig. 3. **Directional task success varies across the eight checkpoint-arena rows.** Points show marginal LEFT and RIGHT success with 95% Wilson intervals. DROID/RoboLab uses 27 matched pairs per row and RoboTwin 63; their tasks and frozen predicates differ. The screen is descriptive, and every causal comparison in this paper is within checkpoint; the figure does not support cross-model or cross-arena ranking.

($p = 1.8 \times 10^{-4}$). On success alone, the layout appears to have no effect; the continuous endpoint measure shows otherwise.

The symmetric-scene cohort contains 4,096 episodes and 2,048 matched pairs across five checkpoint-arena rows. The rows contribute unequally. For $\pi_{0.5}$, 763 pairs comprise 341 at each endpoint plus 27 at each of three intermediate levels; for Nano, 1,123 comprise 521 at each endpoint plus the same three 27-pair levels. Edge, DreamZero, and FastWAM contribute 27 pairs at each endpoint, or 54 each. The intervention sets the registered movable-object asymmetry metric to zero, but it replaces a scene package, including which other objects are present, rather than changing positions alone. Four DROID/RoboLab checkpoints pass the endpoint-redirection positive control and support a scene-effect interpretation. GR00T is outside this cohort because it almost never completes the task in either direction; this is a scope decision, not a failed control. FastWAM and a separate 108-episode LingBot-VA RoboTwin cohort fail the positive control, so we report those rows without treating a smaller gap as a geometry effect. The $\pi_{0.5}$ seeds here are disjoint from the 341 used for reference inversion.

For $\pi_{0.5}$, 341 prompt pairs at each endpoint show the binary RIGHT-minus-LEFT gap falling from 74.2 to 20.2 percentage points and the requested-depth gap from 17.0 to 5.75 cm. At the symmetric endpoint, their paired 90% intervals are [15.0, 25.2] points against a prespecified ± 15.56 -point margin and [4.85, 6.66] cm against ± 4.15 cm; neither is equivalent. The prespecified power audit at $n = 341$ gives an 80%-power minimum detectable effect of 7.8 percentage points for the binary gap, relative to the 15.56-point equivalence margin. A separately registered matched core uses 27 seeds shared across layouts; it estimates interactions of -51.9 points (95% CI $[-81.5, -22.2]$, $p = .00786$) and -12.4 cm (95% CI $[-19.7, -4.9]$, $p = .00348$), respectively. Base-frame redirection at the symmetric endpoint is $+23.9$ cm, with 335/341 positive pairs.

The other positive-control-passing checkpoints show the effect’s cross-checkpoint heterogeneity (Fig. 5). DreamZero’s observed binary gap changes from $+51.9$ to -3.7 points (interaction -55.6 points, $p = .00085$), bringing the measured gap near zero without establishing equivalence. Edge is

underpowered for its -14.8 -point binary interaction, whose 95% interval $[-40.7, +7.4]$ includes zero, while its depth interaction is -19.2 cm (95% CI $[-26.0, -11.5]$).

Failure labels also change descriptively across the full $\pi_{0.5}$ endpoint cohorts, pooling directions within 682 episodes per layout. The control endpoint has 1 pick, 267 transport, 41 wrong-side, and 2 release failures; the symmetric endpoint has 0, 158, 7, and 0, respectively, for 311 versus 165 total failures. These unpaired endpoint totals are descriptive bookkeeping, not a separate interaction test.

VII. DISCUSSION

A binary score can fail in opposite directions. A single binary contrast permits two opposite errors of interpretation. A *false failure* occurs when a direction token redirects behavior, but the sampled scene makes one requested placement harder to complete. Reflection directly demonstrates this possibility; displacement, start-side, role-swap, and symmetric-scene cohorts support it under related interventions. A *false pass* occurs when relation-consistent behavior under a familiar base-frame construction is treated as evidence of grounding even though the same physical goal does not transfer under reference inversion. There, relation preservation and token mirroring predict opposite-signed endpoint contrasts, and the observed inverted contrast is near zero. The four measurements and the two controls distinguish these cases because they fail in different places: a ceiling can hide a large change in placement quality, and an executed trace can change without expressing the intended relation. The frozen predicate remains the correct final test of whether the robot completed the task. It cannot, on its own, say why.

A directional gap belongs to the policy-layout pair. Mirroring the movable objects reverses requested-side depth in all three reflection checkpoints and inverts $\pi_{0.5}$ completion. Across the seven-position sweep, the same contrast depends on reference-object position, and the symmetric scene reduces the requested-side-depth disparity in every checkpoint that still redirects on command. Within each comparison, the evaluated policy, prompts, and seeds are unchanged. The score is therefore not a layout-invariant checkpoint property. The surviving $\pi_{0.5}$

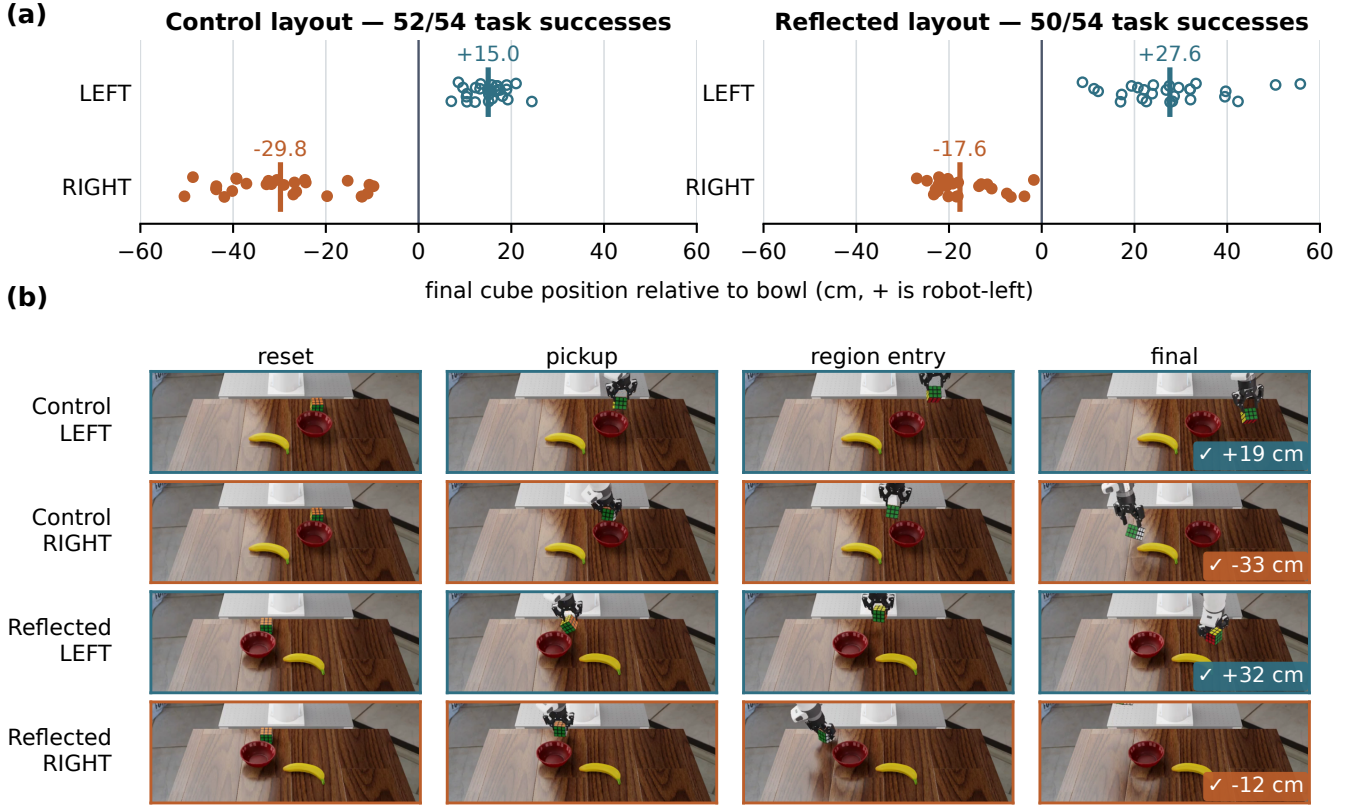


Fig. 4. **Reflecting movable-object positions moves the placement while the binary score stays at ceiling (Nano, 108 episodes).** (a) Final lateral cube position for every seed in the four conditions; bars mark their means and titles give task completion. The prompts still separate after reflection, by 45.3 cm against 44.8 cm in the control layout, but both distributions shift by about 12 cm, which is the requested-side depth interaction of -24.8 cm reported below. Completion moves only from 52/54 to 50/54. (b) The executed rollout for the lowest released seed in each condition, sampled at that episode’s logged events, with ticks marking the frozen predicate and numbers repeating the final signed offset. Robot-left appears toward the image right in this viewport, so the signed values rather than the images fix direction. Rollouts were selected by lowest released seed before any outcome was read.

residual also shows why the conclusion is about the policy–layout pair rather than layout alone.

The controls discriminate rather than explain away. Under the symmetric scene, a powered $\pi_{0.5}$ -specific residual of 20.2 percentage points remains. Its paired 90% interval $[15.0, 25.2]$ excludes zero and is not contained in the prespecified ± 15.56 -point margin, so the residual is separated from zero and also fails equivalence. Scene controls that dissolved every directional gap would be indistinguishable from controls that merely degraded the measurement, which makes a surviving gap the more informative outcome. Here, endpoint redirection remains positive, so the control discriminates a smaller residual rather than erasing the behavior being measured. The result does not locate that residual in the policy: factors not equalized by the scene package, such as per-side reachability among the remaining objects, could still produce it.

A minimum directional-grounding protocol. For relational placement, we recommend matched base-frame LEFT/RIGHT prompts under identical reset and policy seeds, paired with action distinctness, a signed endpoint measure, requested-side depth, and the frozen success predicate. Where stochasticity is

material, add same-prompt repeats. Attribution further requires at least one matched scene intervention with a preserved endpoint-redirection positive control and a reference inversion that preserves the physical goal. Claims that an effect disappeared should rest on equivalence tests rather than a nonsignificant difference.

VIII. LIMITATIONS

Our design trades external validity for control. Exact state matching needs a resettable simulator, so hardware contact uncertainty and natural variation in human instructions remain untested. The object-layout conclusions hold only for the DROID/RoboLab rows that pass the endpoint-redirection control; the RoboTwin rows screen behavior, not a cross-arena replication or refutation. The task family is narrow as well: two named objects, one lateral relation, and one reference inversion on one $\pi_{0.5}$ checkpoint.

The interventions show the measured outcome depending on scene configuration without localizing the cause inside the model or robot. Released material fixes each row’s target-domain dataset but not episode membership or caption exposure, so we make no architecture-level or token-frequency

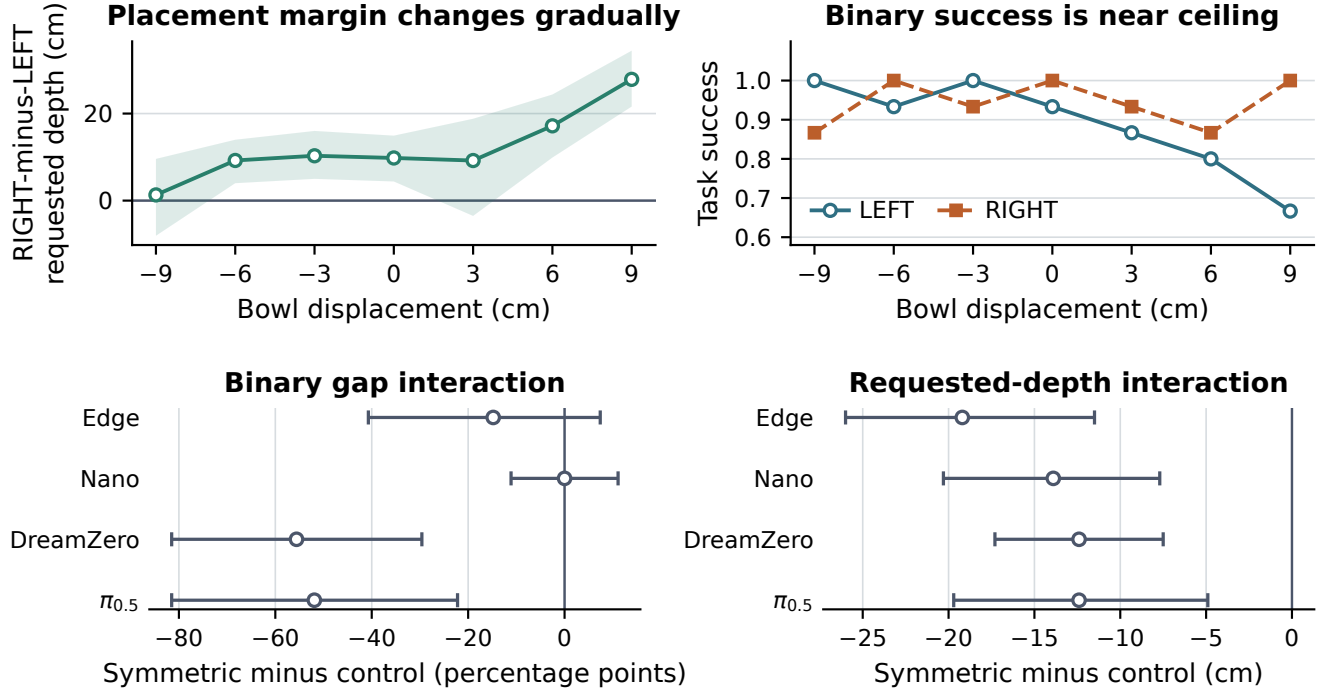


Fig. 5. **Two scene interventions, measured continuously and binarily.** Top: for Nano, the directional placement contrast across the seven-position reference-object sweep, with mean RIGHT-minus-LEFT requested-side depth and seed-clustered 95% intervals on the left and frozen task success by requested direction on the right; each position contains 15 matched seeds. Bottom: symmetric-minus-control interactions in the matched 27-seed core of the positive-control-passing DROID/RoboLab cohort, with seed-clustered 95% intervals; requested-depth disparity decreases in every checkpoint. In both interventions the continuous measure exposes variation that binary success compresses near ceiling, as for Nano. Every comparison is within checkpoint; the panels do not support cross-model ranking.

claim and do not test whether the asymmetry is inherited from training data. If the measured gap ultimately tracks uneven training coverage, the attribution would change but the diagnostic conclusion would not: directional completion would still conflate grounding with coverage-dependent task competence. DreamZero’s public demonstrations do include lateral placements referring to a bowl, so our task may sit closer to its demonstrated set than to the other rows’. Two measurement gaps remain: we did not track reference-object displacement under inversion, though inverted success rates show that the cube still satisfied the placement predicate (Sec. IV), and the fixed-observation diagnostic is action-space only.

IX. CONCLUSION

Reflection reverses completion for $\pi_{0.5}$: 4/27 LEFT and 25/27 RIGHT successes become 25/27 and 9/27 after only the movable-object centers are mirrored in world coordinates, with the policy, prompts, controller, and seeds unchanged. Reference inversion collapses endpoint separation for $\pi_{0.5}$ from 23.28 cm to 0.19 cm on matched physical goals. Ceilings hide placement change: in the matched 27-seed symmetric-scene core, Nano has exactly zero binary interaction ($p = 1$) alongside a -13.9 cm requested-side-depth interaction. These controls expose distinct failures of interpretation: layout can reverse completion, inverted wording can erase familiar-frame separation, and binary success can conceal a continuous placement

change. Directional success therefore does not isolate language grounding. Evaluations should report prompt-conditioned action response, signed endpoint ordering, requested-side depth, and frozen task completion separately, and pair them with matched language and scene controls.

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Short name	Released artifact identifier and revision
$\pi_{0.5}$	pi05_droid_jointpos_polaris v2a010-manifest-f5a56d9565f9381cc
GR00T	nvidia/GR00T-N1.7-DROID 05e7cc97e40dbd33b0890c35cc0214fcb0547ab5
Edge	nvidia/Cosmos3-Edge-Policy-DROID 3ea407af3e156c0af3b4bb6edd85842cc9a58777
Nano	nvidia/Cosmos3-Nano-Policy-DROID 6706d7680581c255ff61e0f3bb49d90eac55c79e
DreamZero	GEAR-Dreams/DreamZero-DROID 96ad344138c66e82536422432ad742f015784942
Efficient-WAM-RT	jiajun0613/Efficient-WAM_RoboTwin/ Efficient-WAM-RT 81280a79e8ac69dd6ff9c9e8698e00d122ec07fd
FastWAM	yuanty/fastwam/ robotwin_uncond_3cam_384.pt 139eebb6d90cdd9bdbbe465f72c6edc9ad5a518a
LingBot-VA	lerobot/lingbot_va_robotwin dle1f93a84eaf9bca9880856fda800cc98cc8eaa robbyant/lingbot-va-posttrain-robotwin 8c9dea8abbc5c91cc9e18bc3264b8915083bbe70

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APPENDIX A SYMMETRIC-SCENE ASYMMETRY METRIC

The registered metric is the root-sum-square departure of the movable objects from bilateral symmetry about the robot midline. It measures midline objects from the midline itself and mirror-paired objects from each other’s reflection, with position weighted at 10 per metre and orientation at 1 per radian. The control layout scores 3.85 and the symmetric layout 0.00, within prespecified millimetre and half-degree tolerances. The robot, cameras, wrist mounting, reset posture, prompts, seeds, and controller are unchanged across the two scenes, but the companion-object inventory changes; the metric therefore certifies movable-object symmetry rather than full-scene or embodiment symmetry.

APPENDIX B EVALUATED CHECKPOINT PROVENANCE

Each row below lists the recorded artifact identifier and revision used in the evaluation. The public audit archive’s manifests contain the available content-hash and runtime metadata, while training-episode membership and caption exposure remain undisclosed; we therefore make no token-frequency or pretraining-data claim.

The $\pi_{0.5}$ row is an internal release rather than a public repository revision, so its recorded identifier is a manifest digest, abbreviated here and given in full in the released manifest. LingBot-VA is evaluated as a base checkpoint plus a RoboTwin post-training checkpoint, so both revisions are required to reproduce it.